

Alternative Week 1 problem session

80-minute small-group class • Problem followed by solution

Teaching assistant deck
Vectors and linear combinations



Suggested rhythm

5 minutes form groups and recall key ideas

8 minutes attempt each problem

3 minutes discuss the solution

10 minutes exit question and summary



Problem 1 Vector calculation

$a=(-1,3,2)$, $b=(4,0,-2)$.

Compute $a+b$ and $3a-b$.

Find the midpoint between a and b .

Check each component.

Work first. The solution is on the next slide.



Solution 1 Vector calculation

$$\mathbf{a} + \mathbf{b} = (3, 3, 0) \quad 3\mathbf{a} - \mathbf{b} = (-7, 9, 8) \quad \text{midpoint} = (3/2, 3/2, 0)$$

1. Use matching components.
2. The midpoint is $(\mathbf{a} + \mathbf{b})/2$.
3. Ask one group to explain the geometry.



Problem 2 Recover vectors

$\mathbf{p} + \mathbf{q} = (5, 7)$ and $\mathbf{p} - \mathbf{q} = (-1, 3)$.

Find $\mathbf{p}$ and $\mathbf{q}$.

Verify both equations.

Work first. The solution is on the next slide.



Solution 2 Recover vectors

$$p=(2,5) \quad q=(3,2)$$

1. Add the given equations to obtain $2p=(4,10)$.
2. Subtract them to obtain $2q=(6,4)$.
3. Verification gives $(5,7)$ and $(-1,3)$.



Problem 3 General combination

$u=(1,2,-1)$, $v=(2,-1,0)$.

Compute $-u+3v$.

Write $cu+dv$ componentwise.

Can $cu+dv$ equal $(5,0,-1)$?

Work first. The solution is on the next slide.



Solution 3 General combination

$$-u+3v=(5,-5,1) \quad cu+dv=(c+2d,2c-d,-c)$$

1. The target third component gives $c=1$.
2. Then $c+2d=5$ gives $d=2$.
3. But $2c-d=0$ also holds.
4. Yes: $u+2v=(5,0,-1)$.



Problem 4 Same direction

$r=(2,-1)$ and $s=(-6,3)$.

Describe all combinations $cr+ds$.

Do the vectors span $\mathbb{R}^2$?

Find a dependence relation.

Work first. The solution is on the next slide.



Solution 4 Same direction

$$s = -3r \quad \text{and} \quad 3r + s = 0$$

1. All combinations equal $(c-3d)r$.
2. They fill only the line through r .
3. They do not span the plane.
4. The nonzero relation proves dependence.



Problem 5 Describe a plane

$v=(1,0,1)$, $w=(1,2,0)$.

Write $cv+dw=(x,y,z)$.

Find one equation connecting x,y,z .

Test $b=(4,6,1)$.

Work first. The solution is on the next slide.



Solution 5 Describe a plane

$$cv+dw=(c+d, 2d, c) \quad \text{so} \quad x=z+y/2$$

1. $z=c$ and $y=2d$.
2. Therefore $x=c+d=z+y/2$.
3. For $b=(4,6,1)$, the right side is $1+3=4$.
4. Thus b belongs to the plane: $c=1, d=3$.



Problem 6 Solve for the weights

Find c, d so that

$$c(1,2)+d(3,-1)=(7,3).$$

Write and solve the component equations.

Work first. The solution is on the next slide.



Solution 6 Solve for the weights

$$c+3d=7 \quad 2c-d=3 \quad \Rightarrow \quad c=16/7, d=11/7$$

1. From the second equation $d=2c-3$.
2. Substitute into the first: $7c-9=7$.
3. Therefore $c=16/7$ and $d=11/7$.
4. Verify both components exactly.



Problem 7 New direction or redundancy

$u=(1,1,0)$, $v=(0,1,1)$, $w=(1,2,1)$.

Does w add a new direction?

Describe the span of u,v,w .

Give a relation that equals zero.

Work first. The solution is on the next slide.



Solution 7 New direction or redundancy

$$w=u+v \quad \Rightarrow \quad u+v-w=0$$

1. w is already a combination of u and v .
2. The three vectors are dependent.
3. Since u and v are not multiples, their span is a plane.
4. Adding w leaves that plane unchanged.



Problem 8 Exit question

In $\mathbb{R}^3$, can two vectors span all of $\mathbb{R}^3$?

Can three vectors span $\mathbb{R}^3$ and still be dependent?

Explain using the idea of new directions.

Work first. The solution is on the next slide.



Solution 8 Exit question

Two vectors give at most a plane. Three dependent vectors also give at most a plane.

1. Each independent vector can add at most one new direction.
2. Reaching $\mathbb{R}^3$ requires three independent directions.
3. If three vectors are dependent, one is redundant and contributes no third direction.